Next.js • Supabase • Gemini AI

SABRATHA UNIVERSITY • FACULTY OF EDUCATION • ENGLISH DEPARTMENT

TEACH-LAP

AI-Powered Teaching Assistant for ELT

Transforming lesson preparation from hours to minutes — private, secure, and tailored to the Standard Classroom

Bilingual Platform • EN / AR • 2025 — 2026

01 DEFINITION

What is TEACH-LAP?

• A private web platform that converts raw texts and documents into structured, academic lectures in seconds

• Powered by Gemini AI — generates: summary (220-350 words), core concepts, key takeaways, classroom applications, scenario Q&A;, reflection tasks & discussion questions

• Every generation is auto-saved to a secure, teacher-isolated history (Supabase) and can be exported as Markdown / .txt

• Bilingual output: English, Arabic, or Bilingual — grounded in SAMR / TPACK and the Libyan Standard Classroom reality

Core Promise

✓ Large classes? Supported.

✓ Limited devices? Optimized for smartphones.

✓ Intermittent internet? Lightweight & fast.

✓ Teacher privacy? 100% isolated spaces.

02 IMPORTANCE

Why It Matters

■ SAVE TIME

Cuts preparation from 3-4 hours to under 5 minutes. Teachers focus on teaching, not formatting.

■ QUALITY & CONSISTENCY

Every lecture follows the same academic rigor: SAMR/TPACK links, ELT skill tags, Libyan context fit.

■ PRIVACY BY DESIGN

No public marketplace. Each teacher sees only her courses, students, and history. Service-role bypass ensures isolation.

■ REAL CLASSROOM FIT

Designed for Standard Classroom — large groups, smartphones only, low bandwidth. No lab required.

Who Is It For?

PRIMARY USERS

- Two dedicated teachers with fully isolated workspaces — each with own login, courses by Year/Semester, and student cohorts
- Teacher A: 4 active subjects (Years 1-4) — e.g., CALL, Conversation, ELT Methodology, Phonology
- Teacher B: 3 active subjects (Years 1 & 4) — e.g., Writing, Literature & Micro-Teaching
- Schedules embedded per subject (Day + Lecture Period + Time)

STUDENTS

- Invited per course/year by their teacher via email — no self-registration chaos
- View published lectures, save summaries (.md/.txt), and see attendance & grades (read-only)
- No Gemini access — consumption and reflection only, keeping assessment fair

MANAGEMENT

- Read-only oversight for quality assurance — no editing of teacher content
- Future-ready: scalable to more teachers without redesign

Benefits & Conclusion

FOR TEACHERS
Generate → Save → Publish → Track. Manage attendance, coursework & exam scores per course. History with search, view, delete & export.

FOR STUDENTS
One private space: My Courses → Lectures (with summaries) → My Grades (attendance + scores). Simple, mobile-first.

FOR THE INSTITUTION
Standardized digital archive, bilingual capability, secure Supabase RLS + service-role isolation, 90-day persistent login.

CONCLUSION

TEACH-LAP is not a generic public site that gets neglected. It is a private, professional hub that solves a real daily problem — preparation time and classroom relevance — while respecting teacher privacy and student fairness. It turns a manual burden into a 5-minute professional workflow.

How It Works — End to End

1. LOGIN

Teacher logs in with username + permanent code → persistent HMAC-signed cookie (90 days) → redirected to her private dashboard (/teacher/teacher-a).

2. COURSES

Courses auto-seeded per subject/year via SQL (teacher_username isolated). Each course has its own interactive page (lectures, students, records).

3. INVITE

Teacher invites students per course/year by email → course_enrollments created → students see only their teacher's courses.

4. GENERATE

Teacher pastes text / uploads files → POST /api/generate → Gemini (gemini-3.5-flash-lite + fallbacks) → JSON → Markdown → saved to lectures_history with teacher_username.

5. HISTORY

GET /api/history (teacher_username filtered, bypass RLS) → searchable history panel (total count, view, expand, delete, export .md/.txt). No Unauthorized.

6. PUBLISH

Teacher selects lecture from history → POST /api/courses/[id]/lectures → course_lectures junction → visible instantly to enrolled students.

7. TRACK

Record attendance (present/absent/late/excused), coursework & exams per course → students see grades read-only in /student.

8. LOGOUT

Header + Dashboard logout clears both teacher_session & student_session (secure/sameSite) → redirects to correct login.

Technical Concept

STACK

• Frontend: Next.js 14 App Router + Tailwind + i18n (en/ar, RTL aware)

• Backend: Supabase (Postgres + Auth + RLS) • Service-role bypass for teacher_session

• AI: Google Gemini API (server-side GEMINI_API_KEY, JSON schema + Markdown render)

• Auth: Custom HMAC teacher_session (username.code) — no Supabase Auth dependency for teachers

DATA & SECURITY

• Tables: courses (teacher_username isolated), course_enrollments, course_lectures, lectures_history (teacher_username + professor_id), attendance, coursework_grades, exam_scores

• RLS policies + is_enrolled_in_course() / is_teacher_of_course() helpers — bypass only for verified teacher_username

• Middleware: protects /teacher, /admin, /courses/[id] — allows /api/auth/* public

• Build: 15 routes + Middleware 57.1kB — verified production build

Setup: 1) Run whitelist.sql → lectures_history.sql → step5_lms.sql 2) Run enas_teacher_setup.sql & eman_teacher_setup.sql 3) Run fix_lectures_history_teacher_username.sql 4) Set TEACHER_SESSION_SECRET + GEMINI_API_KEY on Vercel

07 LIVE PREVIEW

Screens & Experience

/ Home

Hero: Generate lecture form (title + text + files) → Gemini settings (language) → Live Markdown preview + Export

/teacher/[username]

Dashboard: welcome banner, subjects by semester with schedule chips, created courses grid, invite panel, quick tools

/courses/[id]

Course page: 5 tabs — Lectures (publish/unpublish), Students (enroll/remove), Attendance, Coursework, Exams

/student & Header

Student view: My Courses with read-only lectures + grades. Global header: role-aware logout (teacher/student)

PROJECT CREDIT

Technical Project Idea

A. Inas Mesbah

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Conceived, directed and supervised the pedagogical model, data structure and AI workflow for TEACH-LAP.
Private instance: 2 teachers, isolated workspaces, bilingual AI grounded in the Standard Classroom.

This presentation contains no credentials. Access is granted individually via username + permanent code.